25901737 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCMBG101 - MICROBIAL PHYSIOLOGY AND FOOD MICROBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. Which of the following is commonly used as carbon source by microorganisms? [K] / [CO1]

   a). Glucose b). Nitrate

   c). Aminoacids d). None of these

2. Importance of adding PH indicator to enrichment medium. [E] / [CO2]

   a). Used to isolate specific microorganisms b). To inhibit the growth of unwanted microbes c). To indicate changes in PH d). To provide nutrients

3. Identify the physical consistency of nutrient broth. [A] / [CO2]

   a). Liquid medium b). Solid medium

   c). Semisolid medium d). None of these

4. Which of the following is not a component of PDA? [K] / [CO2]

a). Potato b). Dextrose

c). Agar d). Ammonia

5. Write the binomial of bread mold. [K] / [CO3]

a). Aspergillus oryzae b). Rhizopus stolonifer

c). Penicillium expansum d). Mucor racemosus

6. What is/are the main products of fat hydrolysis by lipases? [U] / [CO3]

a). Peptides and amino acids b). Sulfides and mercaptans

c). Fatty acids and glycerol d). Carbon dioxide and water

7. The decay or decomposition or an undesirable nature of food is called ……. [U] / [CO3]

a). spoilage b). contamination

c). cross-contamination d). rotting

8. Choose the pathogen that is spread through raw milk. [U] / [CO4]

a). Salmonella b). Listeria

c). E. coli d). all of the above

9. Which of the following preservatives is commonly used in the preservation of dried fruits? [K] / [CO4]

a). Sorbic acid b). Sodium benzoate

c). Sulfur dioxide d). Acetic acid

10. The chemical preservative that is commonly used in brines and curing solutions for meat and fish. [K] / [CO4]

a). vinegar b). sodium nitrate

c). salt d). All of these

[1x10 = 10]

Part B

Fill In The Blanks

Answer all questions

Each question carries **1** marks

11. Rapid multiplication of bacterial cells observed in _____ of growth. [U] / [CO1]

12. _____ is an example of selective media. [K] / [CO2]

13. Peptone is used as source of _____ in bacteriological media. [U] / [CO2]

14. Foods that spoil readily if special preservative methods are not used are called …….. [U] / [CO3]

15. ……..usually refers to freezing in air at a temperature of −15 to −29 °C. [A] / [CO4]

[1x5 = 5]

Part C

State Whether True or False

Answer all questions

Each question carries **1** marks

16. Thermophiles grow at very low temperatures. [K] / [CO1]

17. Agar agar is used in the preparation of nutrient broth. [U] / [CO2]

18. Water activity of pure water is 1. [K] / [CO3]

19. Temperature is not an intrinsic factor affecting growth of microorganisms in food. [K] / [CO3]

20. The use of multiple hurdles in milder conditions is ineffective in controlling the growth of microorganisms in food. [An] / [CO4]

[1x5 = 5]

Part D

Very Short Answer Type Questions

Answer any **10** questions

Each question carries **1** marks

21. What are chemolithorophs? [K] / [CO1]

22. What is PDA? [K] / [CO2]

23. What are aerobes? [K] / [CO2]

24. Explain the use of peptone? [U] / [CO2]

25. What are thermophiles? [K] / [CO2]

26. Indicate the major kinds of contaminants that enter into food from workers. [U] / [CO3]

27. Give example of a biological leavening agent. [U] / [CO3]

28. Name two inhibitory substances present in food. [K] / [CO3]

29. Give two examples of non-ionizing radiations. [U] / [CO4]

30. Indicate the chief preservation factor/s that prevent spoilage of canned foods. [U] / [CO4]

31. What are the principal bactericidal agents in woodsmoke? [K] / [CO4]

32. Expand GRAS. [K] / [CO4]

[1x10 = 10]

Part E

Short Essay Questions

Answer any **4** questions

Each question carries **3** marks

33. Compare batch culture and continuous culture. [U] / [CO1]

34. Explain spread plate method. [U] / [CO2]

35. Discuss about the significance of any three bacteria importance in food industry. [U] / [CO3]

36. What are the sources of contamination of food from animals? [K] / [CO3]

37. Write a short note on pasteurization and its significance in food preservation. [U] / [CO4]

38. Discuss about the applications of antibiotics as a chemical food preservative. [U] / [CO4]

[3x4 = 12]

Part F

Long Essay Question

Answer any **1** questions

Each question carries **8** marks

39. Design a selective medium that would specifically inhibit the growth of gram positive bacteria. [C] / [CO2]

40. How does low temperature methods preserve food, and what are the key differences between chilling and freezing? [U] / [CO4]

[8x1 = 8]